

# BERKE YAMAN SUNGUR

 [berke-yaman-sungur.github.io/portfolio](https://berke-yaman-sungur.github.io/portfolio) |  [linkedin.com/in/berke-yaman-sungur/](https://linkedin.com/in/berke-yaman-sungur/)

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## EDUCATION

**Boston University**  
**College of Engineering**

**Sep 2026 – Ongoing [Anticipated Graduation Jun 2028]**  
**Boston, MA**

- **MS in Electrical and Computer Engineering**
- **Distinctions:** Admitted with College of Engineering Dean's Graduate Scholarship

**Bilkent University**  
**Faculty of Engineering**

**Sep 2022 – Jun 2026**  
**Ankara, Turkey**

- **BSc in Electrical and Electronics Engineering;** CGPA 3.60/4.00
- **Distinctions:** Comprehensive Scholarship Student and High Honor Student since 2022
  - Achieved the 655<sup>th</sup> ranking in the math-science category and achieved 51<sup>st</sup> ranking in the language category out of approximately 3 million participants in the national university entrance exam in 2022
- **Teaching Assistant (TA) Experience:**
  - EEE202 Circuit Theory: TA responsible for laboratory work and grading (Spring 2025)
  - EEE102 Introduction to Digital Circuit Design: TA responsible for VHDL tutorials (Spring 2024)
  - PHYS101 General Physics I: TA responsible for laboratory work (Fall 2023)
- **Select Projects:**
  - EEE493/494 Capstone Project: [Group Project] Developed a Doppler-based positioning and navigation system using Starlink LEO satellite signals as an alternative to GNSS. Simulated satellite downlinks with SDRs using TLE-derived orbital data and implemented GLRT-based signal acquisition, maximum-likelihood Doppler estimation, Kalman filter tracking, weighted nonlinear least-squares positioning, and Kalman filter navigation, targeting a positioning accuracy of <20 m.
  - EEE485 Statistical Learning and Data Analysis: [Individual Project] Developed an FFT-based audio classification system to identify car engine configurations from idle sounds. Implemented signal preprocessing, spectral feature extraction, and classification using KNN, Linear Regression, and neural networks in MATLAB and Python.
  - EEE412 Microwave Electronics: [Individual Project] Designed and simulated a 1.22 GHz RF amplifier in Keysight ADS, followed by PCB design, component assembly and soldering, RF tuning, and experimental characterization, achieving a measured gain of 12.4 dB.
  - EEE351 Engineering Electromagnetics: [Group Project] Designed and built a functional Rubanoide speaker featuring a flat 100-turn coil (~6  $\Omega$ ) sandwiched between PVC sheets and driven within a uniform magnetic field created by neodymium magnets, applying electromagnetic principles to real-world audio systems through coil design, magnetic field analysis, and electromechanical energy conversion
  - EEE313 Electronic Circuit Design: [Individual Project] Designed, simulated (on LTSpice), and implemented a PCB (using DipTrace) for a temperature control circuit maintaining a resistor at a set temperature, utilizing LM324 op-amps for instrumentation amplification of thermocouple signals and a BC238 NPN transistor as the switching element
  - EEE212 Microcontrollers and Embedded Systems: [Individual Project] Developed a remote-controlled car using NXP FRDM-KL25Z boards and embedded C programming, implementing ADC-based joystick input, UART communication over an RF link with a custom antenna, and PWM motor control for precise speed and steering, gaining hands-on experience in embedded systems, real-time programming, and wireless communication
  - EEE211 Analog Electronics: [Individual Project] Designed a 27 MHz AM RF transceiver with a superheterodyne receiver, and built it, achieving clear two-way voice communication up to 150 m through precise LC filter tuning for frequency stability and low distortion
  - EEE102 Digital Systems Design: [Individual Project] Designed and built a model car that can self-park using the Basys 3 FPGA and VHDL programming
- **Relevant Coursework:** Digital Communications, Microwave Electronics, Statistical Learning and Data Analytics, Principles of Engineering Management, Feedback Control Systems, Engineering Electromagnetics,

Signals and Systems, Electronic Circuit Design, Probability and Statistics, Engineering Mathematics I-II, Circuit Theory, Microcontrollers and Embedded Systems, Analog Electronics, Discrete and Combinatorial Mathematics, Digital Logic Design, Introduction to Programming in Python, Principles of Macroeconomics

Ankara Tevfik Fikret High School

Science/Mathematics Concentration, 100% Merit Scholarship Student

Sep 2018 – Jun 2022

Ankara, Turkey

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## EXPERIENCE

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ASELSAN, Avionics & Guidance Systems

Ankara, Turkey

Paid Part-Time Assistant Engineer (Two Days a Week)

Nov 2025 – May 2026

- Promoted to assistant engineer through past experiences gained, slated to work until the beginning of July 2026
- Continue working on past responsibilities across a broader context

Summer Intern, required by degree

Jun – Jul 2025

- Evaluated the feasibility of interfacing Trusted Platform Module (TPM) units with field-programmable gate arrays (FPGA) via JTAG
- Designed and implemented a hardware security circuit for JTAG inputs, including a complete printed circuit board (PCB) design and layout
- Developed a circuit and PCB to interface a camera module with an SoC test board via the MIPI protocol
- Gained expertise in secure hardware design, high-speed communication protocols, and FPGA-based system integration

Paid Intern (One Day a Week)

Feb – Jun 2025

- Gained hands-on experience designing and implementing communication interfaces such as RS-232, RS-485, RS-422, I<sup>2</sup>C, and USB
- Designed hardware circuits and PCB layouts to realize these communication protocols using Siemens (formerly Mentor) PADS
- Strengthened skills in digital hardware design, embedded communication systems, and PCB development, contributing to advanced avionics solutions

Meteksan Savunma A.Ş., Sensors Systems Engineering

Jun – Jul 2024

Summer Intern, required by degree

Ankara, Turkey

- Fully developed a maritime radar simulation in MATLAB to observe the effects of external factors, such as sea clutter, on a maritime helicopter radar project, which was in early development
- Developed and implemented algorithms to find and track desired targets among the sea clutter in my simulation
- Researched maritime radar technologies and related concepts such as sea clutter, noise, and radar cross-section

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## CLUBS, SKILLS AND HOBBIES

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**International Exam Results:** GRE: Quantitative 170/170, Verbal 155/170 (November 2025)

IELTS Academic: Overall Score 8.0 (November 2025)

**Technical:** Python, C, MATLAB, VHDL, Assembly 8051

**Electronics:** LT Spice, Keysight ADS, Diptrace, Altium, Siemens (formerly Mentor) PADS  
Vivado, Proteus, Keil uVision, Mbed Studio

**Language:** Turkish (Native), English (Advanced), French (Advanced)

**Clubs:** IEEE Bilkent University Student Branch

Bilkent University Bridge Club (Chairman of the Audit Board)

Riders of Bilkent: Bilkent University Motorsports Club (Founding Member)

**Other:** ASELSAN Work Life 101 Program Completion Certificate

**Interests:** Playing the trumpet, music, cars and motorsports, technology, history, traveling